MIL-E-1/78E
4 January 1969
SUPERSEDING
MIL-E-1/78D
21 September 1964

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 5675

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pencil triode, UHF, glass and metal construction

See figure 1

Mounting position: Any

Weight: 1.0 ounce nominal

ABSOLUTE RATINGS: CW oscillator service

Parameter: / Unit:	F1 MHz	Ef V	Eb Vdc	Ec Vdc	Ib mAdc	Ic mAdc	Pp W	tk sec (min)	Rk Ohms	Ehk V	T(seal) °C (note 1)	(E) Barometric pressure, reduced mm Hg
Maximum:	1,700	6.9	300	-90	30	8	5.0	---	---	+90	175	54
Minimum:	---	5.7	---	---	---	---	---	60	---	---	---	---
TEST CONDITIONS:	---	6.3	135	---	---	---	---	300	68	---	---	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) t = 96 hours

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 (see note 2)							
1301	Heater current		0.65	II	If	125	145	mA
1256	Electrode current (1) (anode)		0.65	II	Ib(1)	17	31	mAdc
1306	Transconductance		0.65	II	Sm	5,100	7,700	µmhos
1266	Total grid current	Eb = 150 Vdc; Ec = -2.5 V; Rk = 0	0.65	II	Ic	---	-1.0	µAdc
(E) 1231	Emission	Eb = Ec/Ik = 30 mA; Rk = 0; Ef = 5.5 V	0.65	II	Es	---	14	Vdc
1201	Top shorts	See note 3	0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1236	Power oscillation	Eb = 120 Vdc; Rg/Ib = 25 mA (max); Rk = 0; F = 1.700 ±25 MHz (see note 4)	6.5	S3	Po	300	---	mW
1336	Heater-cathode leakage		6.5	S3	Ihk	---	100	µAdc
1256	Electrode current (2) (anode)	Ec = -25 Vdc; Eb = 120 Vdc	6.5	S3	Ib(2)	---	100	µAdc
1316	Amplification factor		6.5	S3	Mu	15	25	---
1331	Direct-interelectrode capacitance	See note 5	6.5	S3	Cgp Cgk	1.2 2.0	1.6 2.6	pF pF
1031	Low-frequency vibration	Rk = 0; Ec = -2.5 Vdc; Rp = 10.000 ohms; Eb = 150 Vdc	6.5	S3	Ep	---	100	mVac
	Quality conformance inspection, part 3							
(E) ---	Life-test provisions	Group B; Rg/Ib = 30 mAdc; Ic = 15 mAdc (max) (see note 11) TA(seal) = 175° C (min)	---	---	t	500	---	hrs
---	Life-test end point	Power oscillation	---	---	Po	200	---	mW
(E)	Periodic-check tests	See note 6						
1002	Barometric pressure, reduced	Pressure = 55 ±5 mm Hg; voltage = 500 Vac (see notes 6 and 7)	---	---	---	---	---	---
---	Seal fracture	No voltages (see notes 6 and 8)	---	---	---	30	---	lbs
---	Torque	No voltages (see notes 6 and 9)	---	---	---	12.5	---	lb-in.

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Periodic-check tests -Continued							
1116	Lead fatigue	No voltages (see note 6)	---	---	---	---	---	---
1031	High-frequency vibration	No voltages (see notes 6 and 10)	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	See notes 5 and 6	---	---	Cpk	---	0.090	pF

NOTES:

1. Tubes should be provided with adequate cooling to maintain anode seal temperature of less than 175° C.
2. All tests listed under quality conformance inspection, part 1, are to be performed at the conclusion of the holding period.
3. The tube shall be placed in an adapter and mounted on a paddle in accordance with Drawing 267-JAN, or equivalent. The paddle shall be tapped with a rubber tapper (made 30 to 40 durometer rubber) three times in each of two planes, 90° to 120° apart. If a short indication is obtained, the tapping cycle shall be repeated two additional times. If a short indication is again obtained, the tube shall be rejected as inoperable.
4. Tube to be tested in cavity, Drawing 154-JAN, or equivalent.
5. Test in capacitance jig in accordance with Drawing 155-JAN, or equivalent.
6. Tubes for periodic-check tests shall be selected at random at intervals so that when the tube is in production, at the time any lot is offered for acceptance, not less than 6 tubes shall have been checked in the last 6 calendar months. Should the tubes fail a periodic check test, that test shall become a quality conformance inspection, part 2 test (6.5 percent AQL, inspection level S3) until three consecutive lots pass. Thereafter the test will again become a periodic-check test.
7. The specified voltage is to be applied between the anode and the grid plus cathode.
8. The tube shall be supported at the cathode and anode sleeves at points 15/16 ± 1/64 (23.81 mm ± .40 mm) inch apart with the grid disc centered. The force shall be applied to the grid disc, without shock, perpendicular to the axis of the tube. The tube shall not fracture. This is a destructive test. Electrical rejects may be used.
9. The cathode terminal shall be rigidly clamped and a torque of 12.5 lb-in. shall be applied gradually about the tube axis to the anode terminal. The tube shall not fracture. This is a destructive test. Electrical rejects may be used.
10. At the conclusion of the vibration period, the tube shall meet the specified limits for + Ihk, Ib(2), Sm, and low-frequency vibration.
11. Life test in self-excited oscillator cavities, continuous cycle. Alternate conditions:

    (a) Eb = 150 Vdc; F = 1,700 ± 25 MHz
    (b) Eb = 300 Vdc; F = 500 ± 15 MHz

The margins of this tube specification sheet are marked with a revision letter (E) to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:
Army - EL
Navy - EC
Air Force - 11

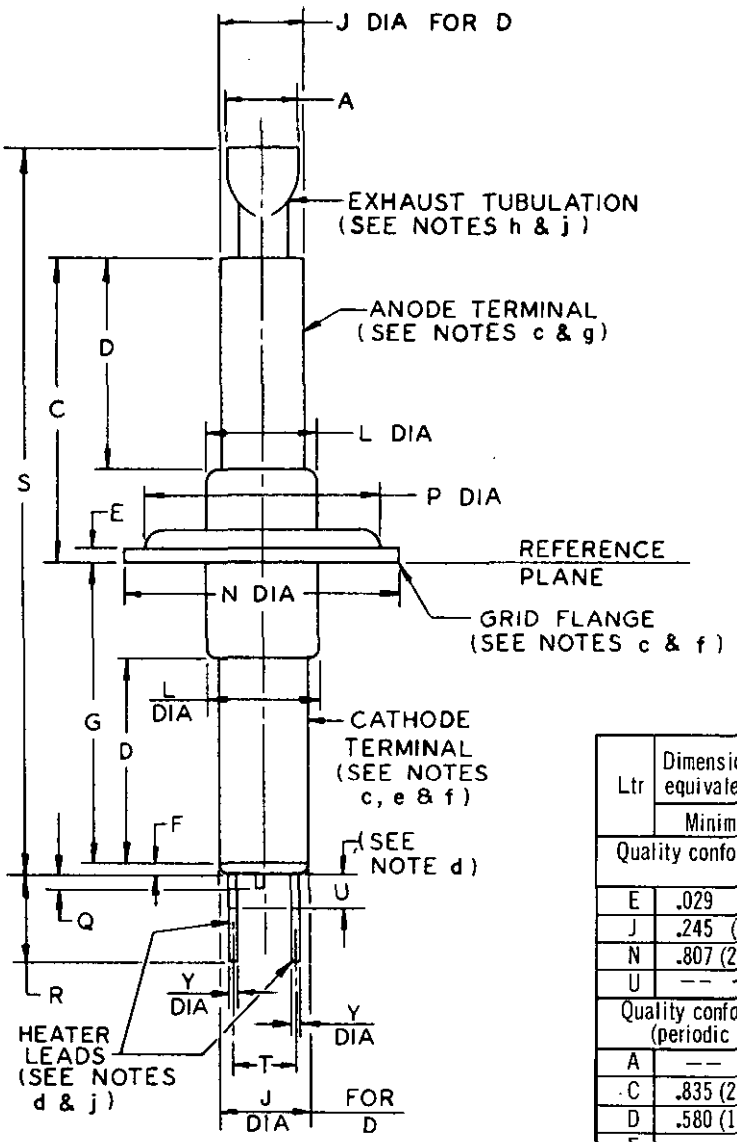
Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Army - EL

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2 (see note a)		
E	.029 (.74)	.035 (.88)
J	.245 (6.22)	.255 (6.47)
N	.807 (20.50)	.817 (20.75)
U	-- --	.080 (2.03)
Quality conformance inspection, part 3 (periodic check, see note b)		
A	-- --	.230 (5.84)
C	.835 (21.21)	.885 (22.47)
D	.580 (14.73)	-- --
F	-- --	.025 (.64)
G	.768 (19.51)	.818 (20.77)
L	-- --	.330 (8.38)
P	-- --	.687 (17.44)
Q	-- --	.040 (1.01)
R	.200 (5.08)	.240 (6.09)
S	-- --	1.981 (50.32)
T	.075 (1.91)	.155 (3.93)
Y	.018 (.46)	.022 (.55)

FIGURE 1. Outline drawing of electron tube type 5675.

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NOTES:

a. These dimensions shall be inspected on a lot-by-lot basis to an AQL of 2.5 percent individual and 4.0 percent combined, sample size code G.

b. These dimensions shall be inspected at the start of production and periodically every 3 months during production. The sample size shall be six tubes and the acceptance number shall be zero defects for each dimension. Failure of any dimension(s) to meet the quality conformance inspection, part 3 requirements for periodic check inspection, shall cause the nonconforming dimension(s) to be inspected on a lot-by-lot basis to an AQL of 2.5 percent individual and 4.0 percent combined, sample size code G. When three successive samples meet all quality conformance inspection requirements for lot-by-lot inspection as specified above, the dimension(s) shall revert to periodic check inspection.

c. Eccentricity of grid flange and of anode shall be .010 (.25 mm) maximum with respect to cathode terminal. Eccentricity of anode shall be gaged by chucking cathode terminal .050 (1.27 mm) minimum and .100 (2.54 mm) maximum from cathode-glass seal and gaging 1/2 TIR, .080 (2.03 mm) from end of anode terminal. Eccentricity of grid flange shall be gaged by chucking cathode terminal as specified above and gaging 1/2 TIR of edge of grid flange. AQL shall be 6.5 percent, inspection level S3.

d. Leads shall be trimmed, and shall be tinned except as specified by dimension U. Untinned length of leads shall be measured from end of cathode terminal.

e. Glass from stem sealing shall not be present on cathode terminal surface. Glass button shall not extend radially beyond cathode terminal surface.

f. Allowable tilt of grid flange shall be .020 (.51 mm) maximum with respect to axis of cathod terminal and shall be determined by chucking cathode terminal, rotating tube through one complete revolution, and gaging total travel distance of flange parallel with axis of cathode terminal and .020 (.51 mm) from edge of flange. AQL shall be 6.5 percent, inspection level S3.

g. Diameter of anode terminal shall not be increased by solder of melted plating from exhause tubulation.

h. Allowable burned area at exhaust tubulation end of anode terminal shall be .0025 square inch maximum. AQL shall be 6.5 percent, inspection level S3.

j. All external metal surfaces, except leads and exhaust tubulation, shall be silver or gold plated 50 msi minimum. Plating shall not be blistered and plated areas shall exhibit no exposed base metal.

FIGURE 1. Outline drawing of electron tube type 5675 - Continued.

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